



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSC323**

Credit Hours: **3 (2, 1)**

Lab Hours/Week: **3**

Course Title: **Operating Systems**

Lecture Hours/Week: **2**

Pre-Requisites: **None**

Course Objective

- To address the purpose, services and implementation design of operating systems;
- To describe the process concept in OS and how are these processes synchronized, scheduled and managed in case of deadlocks;
- To discuss the techniques of primary and virtual memory management;
- To explain the structure of the file systems, storage and I/O devices;
- To describe security and protection issues in computer systems and their management with OS;
- To use C and UNIX commands to develop various system programs under Linux to make use of OS concepts related to process synchronization, shared memory, mailboxes, file systems, etc.

Course Content

This course introduces the services and functions performed by operating system for smooth and accurate system operations. Topics include: Operating Systems Overview; Device Organization & System Operations; Operating Systems structure; Process Management & scheduling; Process Synchronization; Deadlocks; File systems; Memory Management; Storage Management; and Security & Protection.

Unit wise Major Topics

Unit	Topic	No. of Teaching Hours
1.	Operating Systems: Overview, Purpose & Functionality, Evolution, Types; Computing Environments; Device Organization & System Operations: Interrupts, Dual Mode Execution, Single & Multiuser; Services & System Calls, Shell Management; OS Structuring Techniques & Design Issues; and Introduction to Security & Protection	5
2.	Process Management: Concepts, States, Memory Storage, Context Switching; Operations on Processes; IPC Issues; Threads: Parallelism & Concurrency; Scheduling & Dispatching: Types, and Algorithms; Case Studies.	8
3.	Process Synchronization: Primitives, Critical Section Problem & Solutions; Deadlocks: Characterization, Handling; and Case studies.	6
4.	Memory Management: Physical Memory & Issues; Allocation Techniques: Contiguous & Non-Contiguous; Virtual Memory: Demand Paging, Working Sets, and Thrashing; Case studies.	6
5.	Storage Management: Disk Management, Disk Scheduling, Disk Structure, Swap-Space Management, and File System & Implementation; Case studies.	5
Total Contact Hours		30

Mapping of CLOs and GAs

Sr.#	Unit	Course Learning Outcomes	Blooms	GA
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	#		Taxonomy Learning Level			
CLO's for Theory						
CLO-1	1	Describe the purpose, types, services, and structuring techniques of various operating systems	Understanding	2		
CLO-2	2	Analyze the effectiveness of process and thread management mechanisms, including scheduling algorithms and concurrency control, in modern operating systems.	Analyzing	2,3		
CLO-3	3	Apply synchronization and deadlock handling techniques to solve critical section problems and manage process coordination.	Applying	2,3		
CLO-4	4,5	Analyze memory and storage management strategies to evaluate their impact on system performance and resource utilization.	Analyzing	2,3		
CLO's for Lab						
CLO-5	1	Apply system commands and shell scripting techniques to perform basic process, file, and memory management operations in a UNIX/Linux environment.	Applying	2,5		
CLO-6	2-5	Implement solutions for process scheduling, synchronization, memory allocation, and file management to demonstrate key operating system functionalities.	Applying	2-5		
CLO Assessment Mechanism						
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-
Assignments	Assignment 1	Assignment 2	Assignment 3	Assignment 4	-	LAB Assignments
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	-	Lab Mid Term Exam
Final Term Exam	Final Exam					Final Term Lab Exam
Text and Reference Books						
Textbooks: 1. Operating System Concepts, Silberschatz & Galvin, Addison-Wesley, 2021.						
Reference Book: 2. Operating Systems: Internals and Design Principles, Stallings, W., Pearson, 2017. 3. Modern Operating Systems, Tanenbaum, A. S., Prentice Hall, 2014.						